EARTH AND PLANETARY SCIENCE LETTERS 10 (1971) 375–379. NORTH-HOLLAND PUBLISHING COMPANY

PALEOMAGNETISM OF TERTIARY AND RECENT LAVAS OF ISRAEL*

Amos NUR
Department of Geophysics, Stanford University, Stanford, California 94305, USA

Charles E. HELSLEY
Geosciences Division, University of Texas at Dallas, Dallas, Texas 75230, USA

Received 28 September 1970
Revised version received 8 December 1970

Magnetizations in 24 flows of Tertiary age in Israel indicate two stable directions, each of which has both normal and reversed polarities. AF demagnetization decreases the scatter of the NRM results. Typical Tertiary poles are near 70°N 110°W and another set of anomalous poles are near 34°N 50°W. These are similar to other reported Tertiary and Cretaceous poles.

1. Geology and sampling

The sequence of Tertiary and Pleistocene volcanic flows in northern Israel is part of a large volcanic province, extending from east Turkey to east Africa (N. Schulman, personal communication). The flows exposed in the northern part of the Jordan rift zone (fig. 1) are intercalated with sediments which accumulated in a Tertiary inland basin. The oldest units, of Miocene age, consist of a rhythmic sequence of basaltic flows and sediments, covered with a thick series of flows, the so called "Lower Basalt". The "Lower Basalt", which is intensively dissected by faults and joints, is covered by marine sediments and flows of Pliocene age locally referred to as the "Middle Basalt". These are overlain by the "Cover Basalt", also of Pliocene age. The "Cover Basalt" consists of numerous thin basaltic flows, which in turn are overlain by sediments and more flows of Pleistocene age.

A total of 76 samples were obtained by field drilling methods [1]. Each sample's orientation was determined with a sun compass, with an azimuthal error not exceeding 2°. The relative stratigraphic position of all samples was carefully recorded. Whenever

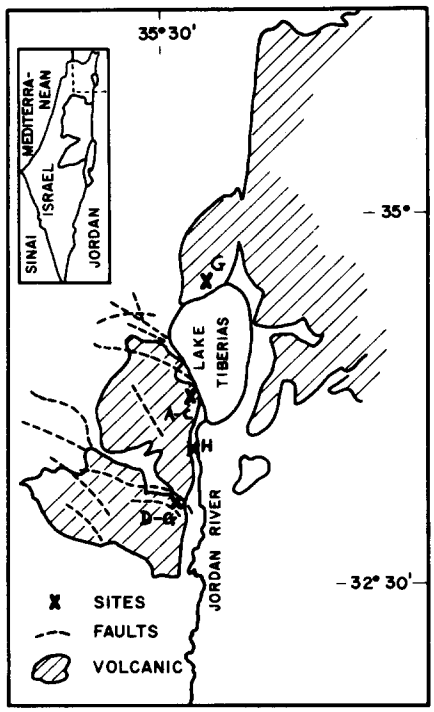


Fig. 1. Map of northern Israel showing the sampling sites.

* University of Texas at Dallas contribution No. 129

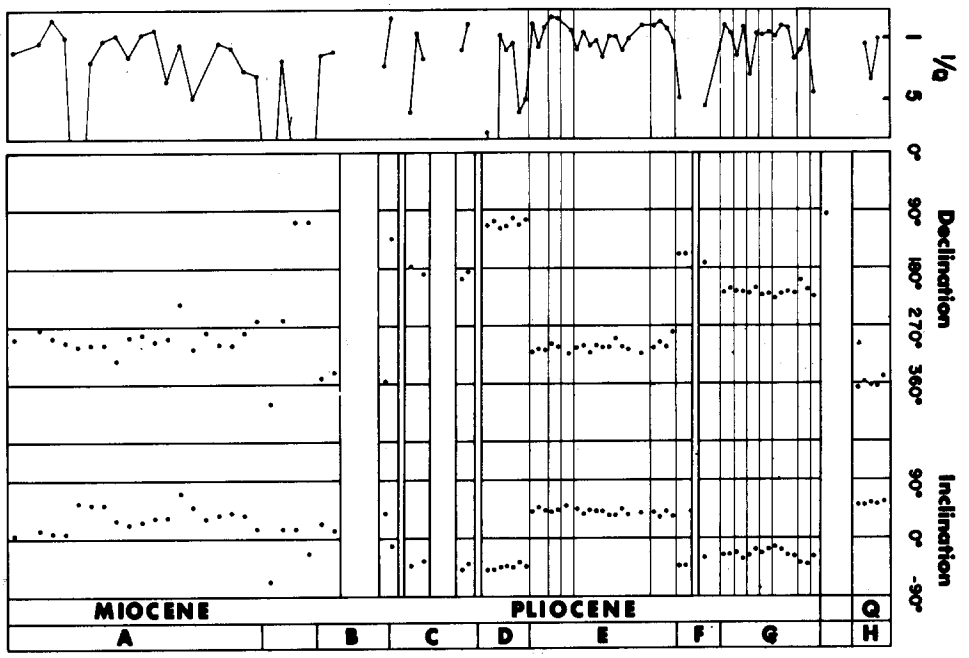


Fig. 2. Direction of magnetization of samples. Samples are ordered stratigraphically. Also shown is the ratio, Q^{-1}, of the magnetic susceptibility to initial intensity of magnetization of the samples. Average pole positions were computed for each group A to H.

possible at least two cores were drilled from each flow.

2. Measurements and computations

Short cylindrical samples, 2.54 cm long and 2.54 cm in diameter, were prepared from the cores, and directions and intensity of magnetization were measured. All samples were demagnetized in an alternating field (AF) until considered stable, usually at peak field intensity of 350–700 oersted. The samples from the "Lower Basalt" showed only small changes in directions of magnetization upon magnetic cleaning – even at demagnetizing fields of 2800 oersted. Values from the other basalts show good clustering upon demagnetization. The declinations and inclinations of all samples, after application of structural corrections and magnetic cleaning are shown in fig. 2. The horizontal thin lines indicate flow boundaries, wherever observable. Note that the section is not to scale in time, and indicates only the stratigraphic sequence of the samples and not the true thicknesses of the units.

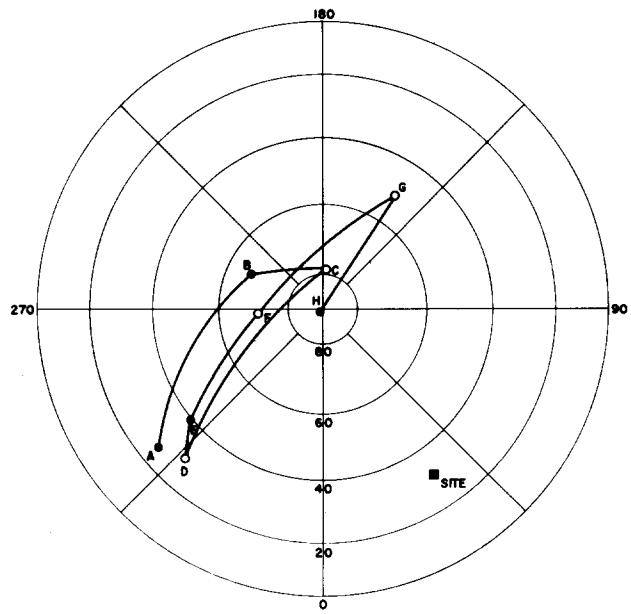


Fig. 3. Paleomagnetic poles for each of the groups of lava flows indicated in fig. 2. The presence of both normal and reversed polarities in each group of poles indicates a high degree of magnetic stability.

Table 1
Average directions of magnetization and the corresponding pole positions.
Samples were grouped according to their magnetization directions.

Sample group	Approx. age	Declination	Inclination	No. of samples	K*	Longitude	Latitude	Polarity
A	Miocene	− 70.2	32.5	19	14	50°W	26°N	normal
B	Pliocene	− 10.9	24.4	3	29	116°W	65°N	normal
C	Pliocene	− 172.3	− 41.7	4	94	86°E	79°N	reversed
D	Pliocene	111.6	−44.7	7	192	42°W	31°N	reversed
E	Pliocene	− 57.4	41.3	20	101	50°W	39°N	normal
F	Pliocene	163.2	− 38.9	3	67	87°W	72°N	reversed
G	Pliocene	− 143.5	− 25.4	15	76	147°E	51°N	reversed
H	Quaternary	− 1.1	52.8	5	319	34°W	89°N	normal

* K is Fisher's precision parameter.

The bulk magnetic susceptibility of all samples was also measured and the Königsberg ratio Q was computed. The inverse values of Q, Q^{-1}, was also shown in fig. 2, in order to emphasize several sharp low values which remain unexplained.

The samples have been separated into eight groups on the basis of changes in declination and inclination. Mean pole positions were computed for each of the various groups of samples, assuming a dipolar field, and these are summarized in fig. 3 (with 4 omitted

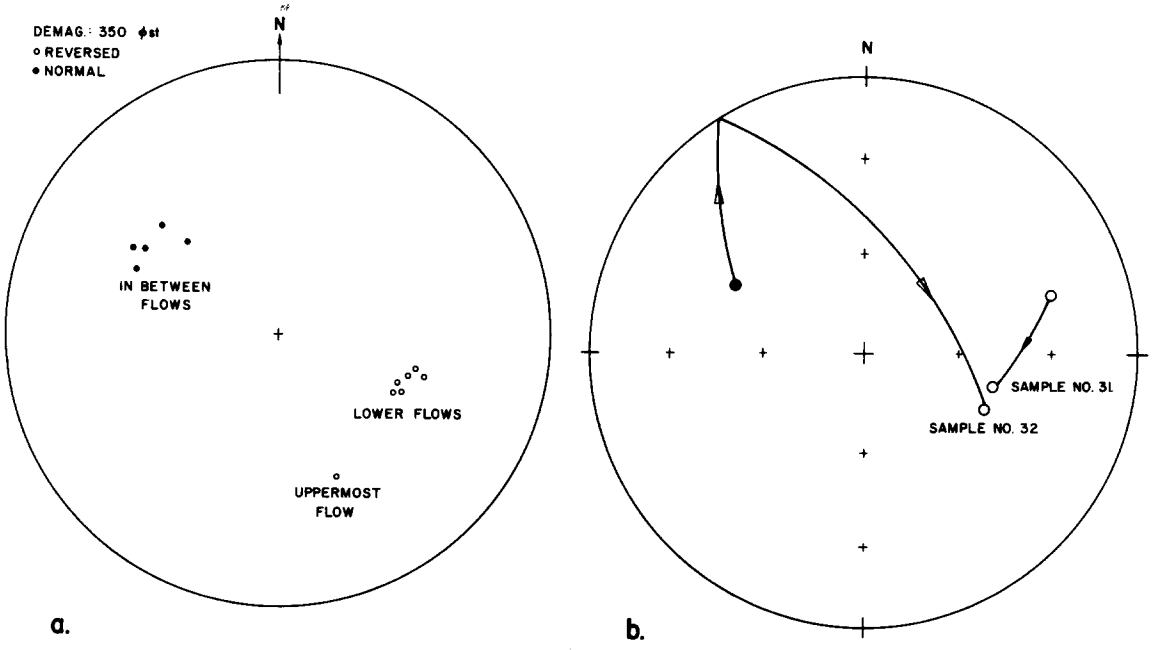


Fig. 4. Evidence for magnetic stability. (a) Representative directions for samples from a continuous series of flows after AC cleaning. (b) Two samples from a single flow before and after AC cleaning (to 350 oersted).

Table 2
Cretaceous and Tertiary pole positions from the Near East.

Reference	Age	Site	Longitude	Latitude	Polarity
Van derVoo [4]	Eocene	Turkey	66°W	65°N	N + R
	Upper Cretaceous	Turkey	99°W	70°N	
Van Dongen [3]	Pliocene	Syria	71°W	73°N	N + R
Helsley and Nur [5]	Upper Cretaceous	Israel	96°W	42°N	N
New results	Miocene and Pliocene	Israel	50°W	34°N	N + R

samples, between 13 and *C*, because of scatter) and table 1.

3. Poles positions

It is immediately apparent in fig. 3 and table 1 that at least two average directions of magnetization are present. These two directions change back and forth with time and involve both normal and reversed polarities. Are these two directions representative of an ancient magnetic field? The presence of both normal and reversed polarities in our samples, and the convergence upon demagnetization (fig. 4), both indicate a high degree of magnetic stability. All samples, except for the uppermost Pliocene (G, fig. 1) and the Pleistocene (H) ones came from essentially one structural block. It is highly unlikely, considering the geology of the area, that any unrecognized differential rotations about a vertical axis (of the order of 60°) or tilting in this block took place.

Pole position G is singularly different from all other positions. Since the samples came from a separate structural block, it is possible that structural effects which could not be established in the field are responsible for this difference.

Thus the presence of two groups of pole positions as shown in fig. 3, each with normal and reversed polarity, indicates that the magnetic field very likely had two relatively "stable" directions. Similar fluctuations in pole positions have been reported for Tertiary rocks in Oregon by Watkins [2]. Pole positions for Syria (Van Dongen et al. [3]) and Turkey (Van der Voo [4]) are given for comparison in table 2.

4. Conclusions

At least four geomagnetic reversals have been recorded in volcanic rocks in northern Israel. Some of the paleomagnetic pole positions computed from the measured samples are close to values reported for Tertiary rocks in other areas. Other portions of the results suggest an anomalous direction which is similar to other Tertiary and Cretaceous anomalous results from Europe and North America.

Acknowledgements

We wish to extend our gratitude to several persons. Dr. N. Schulman and Dr. R. Freund from the Hebrew University, Jerusalem, directed us to the field sites and helped to collect the samples. Dr. Wulf Gose read the manuscript and made valuable suggestions. This project was supported by the National Aeronautics and Space Administration through grant NGL-44-004-001 and by the National Science Foundation through grants GP-2250, GA-590, and GA-1304.

References

[1] C.E. Helsley, Advantage of field drilling of samples for paleomagnetic studies, in: Methods of Paleomagnetism, eds. D.W. Collinson, K.M. Creer and S.K. Runcorn (Elsevier, Amsterdam, 1967).
[2] N.D. Watkins, Paleomagnetism of the Columbia plateaus, J. Geophys. Res. 70, 6 (1965) 1379.
[3] P.G. Van Dongen, R. Van derVoo and T. Raven, Paleomagnetic research in the central Lebanon mountains and the tartous area, Tectonophysics, 4, 1 (1967) 35.

[4] R. Van der Voo, Jurassic, Cretaceous and Eocene pole positions from northeastern Turkey, Tectonophysics 6, 3 (1968) 251.

[5] C.E. Helsley and A. Nur, The paleomagnetism of Cretaceous rocks from Israel, Earth Planet. Sci. Letters 8 (1970) 403.